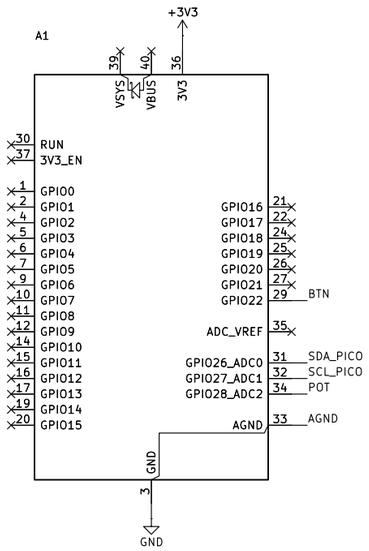
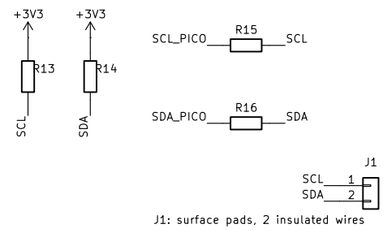
1. Microcontroller: Raspberry Pi Pico (castellated)

I2C1: SDA = GP26, SCL = GP27 (Wire1). Button = GP22.
Potentiometer = GP28 (ADC2). Build the firmware with FRUTAMIDI_PCB.



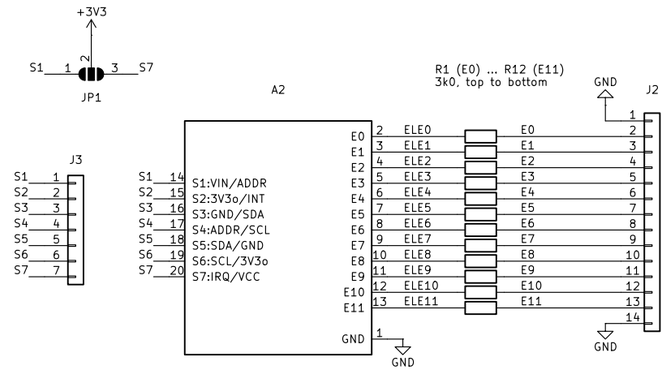
2. I2C bus

R13/R14: 4.7k pull-ups to 3V3 (rise time at 400 kHz, T-050).
R15/R16: 220R between the Pico pins and the bus (wrong-wire current limit, T-055).



J1: surface pads, 2 insulated wires

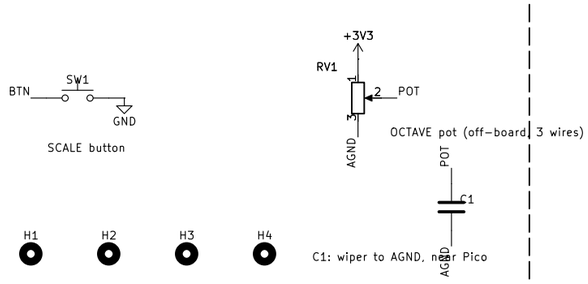
3. Touch sensing: MPR121 module, electrode resistors, electrode wire pads



Module configuration (component side):
  Original module (no STEMMA QT): JP1 bridge 'ORIG' (3V3 -> S1)   wires SDA -> S5   SCL -> S6
  STEMMA QT module (current):   JP1 bridge 'QT'   (3V3 -> S7)   wires SDA -> S3   SCL -> S4
Module GND enters through the electrode row (pin 1), the same on both revisions.

R1-R12: 3.0k in series with each electrode (ESD current limit, ADR-023);
pulse-rated HPWC2512, land gap 5.0 mm, no track under the body.

4. User controls and mounting holes



Pico + MPR121 (1982), single-sided SMD PCB, copper on component side
github.com/catuta0/fruta-midi-controller

Sheet: /
File: frutamidi.kicad_sch

Title: FrutaMIDI — capacitive MIDI controller

Size: A3	Date: 2026-11-09	**Rev: 1.3**
KiCad E.D.A. 10.0.4		Id: 1/1